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2024)**
**The Brexit Vote, Productivity Growth, and Macroeconomic Adjustments in
the U.K.**

Contents

1	Big picture	3
2	Data and measurement: detailed, lightly concise construction	3
2.1	Post-referendum U.K. facts and normalization	3
2.2	Cross-country productivity data for EU accession evidence	4
2.3	U.K. tradable and non-tradable sectoral data	4
2.4	Table 2: industry shares and the meaning of tradability	4
2.5	Table 3: observables used for model estimation	4
2.6	Measurement choices for expenditure ratios and post-vote comparison	5
3	Models and empirical specifications	5
3.1	Sectoral production and factor specificity	5
3.2	Stationary TFP shocks and trend-growth shocks	5
3.3	Sectoral productivity growth and balanced growth	6
3.4	Households, labor disutility, and CES consumption/investment aggregators	6
3.5	External borrowing, domestic bonds, and interest rates	6
3.6	Trade balance and resource constraints	6
3.7	Brexit news-shock specification	7
4	Identification and empirical design: detailed but concise strategy	7
4.1	Overview: mechanism identification rather than one causal coefficient	7
4.2	Step 1: U.K. event facts discipline the shock	7
4.3	Step 2: EU-accession evidence supports the sign of the productivity news	7
4.4	Step 3: pre-referendum structural estimation	8
4.5	Step 4: Brexit shock calibration and out-of-target validation	8
4.6	Step 5: alternative shocks and threats	8
5	Tables and figures: names, reads, interpretations, and conclusions	8
5.1	Figure 1: Adjustments of the U.K. economy following the Brexit vote	8
5.2	Figure 2: Mean productivity growth difference between tradable and non-tradable sectors and Table 1: The effects of joining the EU on productivity growth differentials between sectors . .	9
5.3	Table 2: Industries shares in the non-tradable and tradable sector (%)	10
5.4	Table 3: Observables used for model estimation and Table 4: Calibrated parameter values . .	11
5.5	Table 5: Priors and posteriors for estimated parameters	12
5.6	Figure 3: Main model responses in benchmark Brexit scenario and Figure 4: Additional model responses in benchmark Brexit scenario	13
5.7	Table 6: Comparison of Brexit simulation with U.K. data	14

5.8	Table 7: Estimates of long-run effects of Brexit on U.K. GDP	15
6	Interpretive synthesis and limitations	16
7	Short summary	16
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Final takeaway	17

1 Big picture

- **Question.** How should we understand the U.K. macroeconomic adjustment after the June 2016 Brexit referendum, especially the combination of lower aggregate demand, resilient tradable output, weak non-tradable output, and a persistent fall in the relative price of non-tradables?
- **Core answer.** The paper interprets the referendum as *news* about a future slowdown in tradable-sector productivity growth, rather than as an immediate realized productivity collapse.
- **Why the distinction matters.** A future productivity slowdown affects expectations immediately. Relative prices, consumption, investment, and desired sectoral allocation respond before the productivity slowdown is actually realized.
- **Key empirical facts.** After the vote, aggregate GDP growth slowed, tradable GVA remained relatively resilient, non-tradable GVA stagnated relative to trend, and the relative price of non-tradable GVA fell and then grew along a lower parallel path.
- **Mechanism.** Anticipated lower future productivity in tradables lowers the expected future relative value of tradable production. The current relative price of non-tradables falls. Before the productivity decline arrives, tradable producers benefit from relatively favorable current production conditions - the paper's "sweet spot" effect.
- **Sectoral reallocation.** The tradable sector expands in the short run, while the non-tradable sector contracts. Aggregate activity still weakens because households and firms anticipate lower future income and reduce current consumption and investment.
- **Data contribution.** The paper constructs a U.K. quarterly macro data set splitting GVA, hours, labor productivity, and relative prices into tradable and non-tradable sectors using 2-digit SIC industry information.
- **External evidence.** Cross-country evidence from EU accession episodes shows that joining the EU is associated with faster productivity growth in tradable sectors relative to non-tradable sectors. This motivates the Brexit shock as news about lower future tradable productivity growth.
- **Structural contribution.** The authors estimate a real two-sector small open economy model using pre-referendum U.K. data, then feed in a Brexit news shock after 2016:Q2. The shock is not used to estimate model parameters.
- **Quantitative result.** The simulation matches the direction of most sectoral and aggregate post-referendum movements. It implies a long-run GDP reduction of about 3.6%, at the low end of existing Brexit estimates.
- **Main takeaway.** Bad long-run news about tradable productivity can temporarily support tradable activity through relative-price adjustment, even while aggregate consumption, investment, and long-run output fall.

2 Data and measurement: detailed, lightly concise construction

2.1 Post-referendum U.K. facts and normalization

- Figure 1 uses $100 \times \log$ indexes normalized to 2016:Q2, the referendum quarter.
- Panel A compares aggregate GDP, tradable real GVA, and non-tradable real GVA; Panel B plots the relative price of non-tradable GVA relative to tradable GVA.
- The key empirical pattern is sectoral: aggregate GDP slows, tradables remain relatively resilient, non-tradables stagnate, and the relative price of non-tradables shifts down.

Interpretation: These facts discipline the mechanism. A one-sector recession story can explain slower GDP, but not the joint movement of tradable output, non-tradable output, and the internal relative price.

2.2 Cross-country productivity data for EU accession evidence

- The cross-country exercise uses OECD and Eurostat quarterly national accounts for 31 countries and 10 sectors over 1997:Q1–2016:Q2.
- The dependent variable is quarterly growth in real output per hour at the country-sector-quarter level.
- Fifteen countries join the EU during the sample, creating staggered accession variation.
- The baseline classification treats agriculture, manufacturing, and financial services as tradable; construction and other service sectors are non-tradable.
- Growth rates are winsorized, and some countries use employees rather than total hours where hours are unavailable.

Interpretation: This is not a direct Brexit estimate. It supports the sign of the shock: EU integration raises tradable-sector productivity growth relative to non-tradables, so Brexit is plausibly adverse news for future tradable productivity.

2.3 U.K. tradable and non-tradable sectoral data

- The authors build a U.K. quarterly sectoral dataset from ONS 2-digit SIC industries over the pre-referendum period.
- Industries are classified as tradable when more than 10% of final demand is traded.
- They aggregate real GVA, nominal GVA, implicit deflators, hours, and labor productivity into tradable and non-tradable sectors.
- Sectoral labor productivity is real GVA divided by hours; the relative price is the non-tradable GVA deflator divided by the tradable GVA deflator.
- The resulting hours split is close to 45% tradable and 55% non-tradable, which helps calibrate labor-disutility weights.

Interpretation: The data construction is central because the model requires sectoral quantities, hours, productivity, and relative prices that are not available directly in standard aggregate accounts.

2.4 Table 2: industry shares and the meaning of tradability

- Table 2 shows that services dominate both sectors: 73.50% of tradable GVA and 83.26% of non-tradable GVA.
- Manufacturing is concentrated in tradables, while construction is non-tradable.
- Excluding government-related sectors gives similar non-tradable GVA dynamics, with a correlation of 0.93 relative to the baseline.

Interpretation: Tradables are not just manufacturing. U.K. tradable services are large, so the paper’s Brexit mechanism applies to a broad modern service economy.

2.5 Table 3: observables used for model estimation

- The model is estimated on nine pre-referendum series: consumption/GDP, investment/GDP, hours per capita, sectoral productivity growth, relative-price growth, and two U.S. real interest-rate measures.
- The estimation ends in 2016:Q2, before post-referendum outcomes are used for evaluation.

- The Kalman filter handles different sample start dates; measurement errors are allowed for selected variables.

Interpretation: The model is disciplined by aggregate, sectoral, price, and interest-rate data before Brexit. This avoids fitting structural parameters to the post-vote episode.

2.6 Measurement choices for expenditure ratios and post-vote comparison

- Nominal consumption/GDP and investment/GDP ratios preserve the longer sample because quarterly sectoral deflators begin later.
- Table 6 compares data and model responses using log deviations from a pre-referendum trend estimated over 2000:Q1–2016:Q2, with a post-2010 level-shift dummy.
- The Brexit simulation is calibrated to match the fall in the relative price of non-tradables; other outcomes are out-of-target checks.

Interpretation: These choices make the exercise a disciplined quantitative accounting exercise: target the key relative price, then ask whether quantities, wages, rates, and trade dynamics move consistently.

3 Models and empirical specifications

3.1 Sectoral production and factor specificity

For each sector $M \in \{T, N\}$, firms produce with Cobb-Douglas technology:

$$Y_{M,t} = a_{M,t} K_{M,t}^{\alpha_M} (Z_{M,t} n_{M,t})^{1-\alpha_M}.$$

- $a_{M,t}$ is stationary TFP.
- $Z_{M,t}$ is sector-specific labor-augmenting productivity with stochastic trend growth.
- Labor is sector-specific; capital in each sector is composed of tradables and non-tradables.
- Perfect competition implies zero profits in equilibrium.

Interpretation: The production block is designed to allow sector-specific productivity growth and sector-specific labor reallocation. This is the minimum structure needed to explain why tradables and non-tradables move differently after the Brexit vote.

3.2 Stationary TFP shocks and trend-growth shocks

Stationary sectoral TFP follows:

$$\ln a_{M,t} = \rho_M^a \ln a_{M,t-1} + \varepsilon_{M,t}^a.$$

Trend productivity growth follows:

$$\ln(g_{M,t}^z/\bar{g}_M^z) = \rho_M^g \ln(g_{M,t-1}^z/\bar{g}_M^z) + \varepsilon_{M,t}^g.$$

Interpretation: The distinction between level shocks and growth shocks matters. A shock to growth has permanent level implications, which is why it can capture long-run Brexit costs while still allowing an anticipation phase before the decline materializes.

3.3 Sectoral productivity growth and balanced growth

$$X_{M,t} = (X_{T,t})^{\alpha_M} (Z_{M,t})^{1-\alpha_M}, \quad g_{T,t} = g_{T,t}^z, \quad g_{N,t} = (g_{T,t})^{\alpha_N} (g_{N,t}^z)^{1-\alpha_N}.$$

Interpretation: The non-tradable sector is partly tied to tradable technology through capital composition. This is why a tradable productivity slowdown has long-run implications for non-tradables even though the shock directly hits tradables.

3.4 Households, labor disutility, and CES consumption/investment aggregators

$$U_t = \frac{\left[C_t - X_{T,t-1} \epsilon_t \left(\frac{\theta_T}{\omega_T} n_{T,t}^{\omega_T} + \frac{\theta_N}{\omega_N} n_{N,t}^{\omega_N} \right) \right]^{1-\gamma}}{1-\gamma}.$$

Consumption and investment are CES aggregates of tradable and non-tradable goods:

$$C_t = \left[\zeta^{1+\sigma} C_{T,t}^{-\sigma} + (1-\zeta)^{1+\sigma} \left(\frac{X_{T,t-1}}{X_{N,t-1}} C_{N,t} \right)^{-\sigma} \right]^{-1/\sigma},$$

$$I_t = \left[\zeta^{1+\sigma} I_{T,t}^{-\sigma} + (1-\zeta)^{1+\sigma} \left(\frac{X_{T,t-1}}{X_{N,t-1}} I_{N,t} \right)^{-\sigma} \right]^{-1/\sigma}.$$

Interpretation: The estimated elasticity of substitution is below one, so tradables and non-tradables are gross complements. This complementarity is central: lower expected future income reduces demand for both sectors, reinforcing the fall in the relative price of non-tradables.

3.5 External borrowing, domestic bonds, and interest rates

The external interest rate includes a very small debt-elastic premium and an exogenous foreign-rate shock:

$$r_t^* = \bar{r}^* + \psi \left(\exp(B_{t+1}^*/X_{T,t} - \bar{b}^*) - 1 \right) + (\exp(\mu_t) - 1).$$

Interpretation: The small open economy can smooth intertemporally through foreign borrowing. The authors set the debt-elastic premium to be tiny, so the results are not driven by an endogenous sovereign-risk/default-premium story. The domestic interest-rate response is instead connected to productivity-news expectations and relative-price dynamics.

3.6 Trade balance and resource constraints

The tradable trade balance is:

$$TB_t = B_t^* - \frac{B_{t+1}^*}{1+r_t^*}.$$

Tradable and non-tradable output are allocated across consumption, investment, government demand, adjustment costs, and net exports.

Interpretation: Because only tradables can be exported/imported, the relative-price adjustment after the news shock affects both sectoral production incentives and the trade balance. The paper later notes that the model is better at net-export logic than gross import/export dynamics.

3.7 Brexit news-shock specification

The Brexit experiment adds a separate productivity-growth component not used in pre-referendum estimation. It evolves as:

$$\ln g_{T,t}^B = \vartheta \ln(g_{T,t-1}^B) + (1 - \vartheta) \ln(\bar{g}_T) + \varepsilon_t^B,$$

with slow convergence parameter $\vartheta = 0.95$. A second smoothing equation gives the realized growth process inertia, with smoothing parameter $\varrho = 0.8$.

- In quarter 0, the economy is on its initial balanced growth path.
- In quarter 1, agents learn that Brexit will happen in quarter 15.
- The shock is a negative future shock to tradable productivity growth.
- Agents perfectly foresee the future shock once the news arrives.
- The shock size is calibrated to match the trough decline in the relative price of non-tradables after the vote.

Interpretation: The news shock is deliberately separated from ordinary estimated productivity disturbances. This reflects the authors’ view that Brexit was a rare, salient, forward-looking event rather than a typical business-cycle shock.

4 Identification and empirical design: detailed but concise strategy

4.1 Overview: mechanism identification rather than one causal coefficient

The paper does not estimate one treatment effect of Brexit. It identifies a mechanism: the referendum is modeled as news about a lower future path for tradable productivity growth. The evidence is triangulated from sectoral U.K. facts, EU-entry evidence, pre-Brexit structural estimation, and alternative-shock comparisons.

Interpretation: The identifying claim is not “Brexit caused exactly X.” It is that one disciplined productivity-news channel explains a large share of the observed sectoral adjustment.

4.2 Step 1: U.K. event facts discipline the shock

1. Construct tradable and non-tradable quarterly U.K. series from 2-digit industry data.
2. Normalize around 2016:Q2 and compare post-vote paths with pre-vote trends.
3. Use the joint facts – resilient tradables, weak non-tradables, and a lower relative price of non-tradables – to restrict plausible shocks.

Identification logic: A pure aggregate demand shock, an immediate tradable productivity collapse, or a generic recession would not naturally match this sectoral pattern. The news shock can, because relative prices adjust before future productivity falls.

4.3 Step 2: EU-accession evidence supports the sign of the productivity news

1. Estimate whether productivity growth rises more in tradables after EU entry.
2. Use country, sector, time, country-sector, and country-time fixed effects to absorb broad confounders.
3. Cluster standard errors by country and report robustness checks, including placebo accession dates and alternative tradable definitions.

Identification logic: EU entry is not Brexit, but it supplies external evidence that integration mainly raises tradable productivity growth. Brexit is interpreted as the opposite news.

4.4 Step 3: pre-referendum structural estimation

1. Estimate the two-sector small open economy model using only data through 2016:Q2.
2. Calibrate steady-state targets such as sectoral labor shares, expenditure shares, trend growth, discounting, and debt-premium elasticity.
3. Estimate substitution, labor-supply, adjustment-cost, persistence, volatility, and measurement-error parameters with Bayesian methods.

Identification logic: The model's structure is learned before the referendum shock is evaluated, limiting post-event overfitting.

4.5 Step 4: Brexit shock calibration and out-of-target validation

1. Impose news in 2016:Q2 of a future fall in tradable productivity growth.
2. Set the anticipation horizon to align with the expected Brexit implementation period.
3. Choose the shock size to match the relative price decline of non-tradables.
4. Compare other variables – output, hours, consumption, investment, wages, interest rates, and trade balance – to data at 6- and 12-quarter horizons.

Identification logic: The relative price is the target because it is the key mechanism object. The remaining variables test whether the same shock produces the broader adjustment.

4.6 Step 5: alternative shocks and threats

- A future tradable productivity *level* shock fits some quantities but produces a less persistent interest-rate response.
- A future non-tradable productivity acceleration has little anticipation effect and implies a long-run expansion, which is implausible for Brexit.
- Main threats include uncertainty shocks, gross trade and inventory dynamics, monetary/risk-premium channels, CPI real-exchange-rate movements, and the extrapolation from EU entry to EU exit.

Interpretation: The strategy is convincing because several independent moments line up: U.K. sectoral facts, EU-entry evidence, pre-Brexit model estimation, and alternative-shock failures. It remains a mechanism quantification, not the full total effect of Brexit.

5 Tables and figures: names, reads, interpretations, and conclusions

5.1 Figure 1: Adjustments of the U.K. economy following the Brexit vote

Read:

- Panel A plots aggregate GDP, tradable GVA, and non-tradable GVA as $100 \times \log$ indexes normalized to 2016:Q2.

- GDP grows more slowly after the referendum, but the aggregate path hides a sectoral split.
- Tradable GVA rises relative to the pre-referendum baseline, while non-tradable GVA stagnates.
- Panel B plots the relative price of non-tradable GVA. It falls sharply after 2016:Q2, then resumes a broadly similar growth rate from a lower level.
- The dashed line in Panel B extrapolates the pre-referendum growth trend, making the level shift easy to see.

Interpretation: The figure is the paper’s starting point. If Brexit had simply been an immediate aggregate demand collapse, one would expect both sectors to weaken together. Instead, tradables temporarily look strong while non-tradables weaken, and relative prices move in a way consistent with anticipated future tradable productivity deterioration. The fall in the relative price of non-tradables is especially important because it occurs before the future productivity loss fully materializes.

Conclusion: Figure 1 motivates the entire mechanism: Brexit news creates a short-run reallocative episode, not merely an aggregate recession.

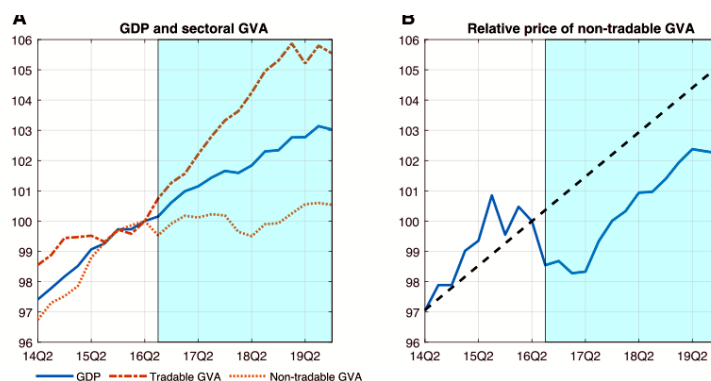


FIGURE 1
Adjustments of the U.K. economy following the Brexit vote
Notes: The unit in both panels is $100 \times \log$ index with 2016:Q2=100. Details on the construction of the underlying sectoral data is provided in Section 4.1.

(Ehali and Warren, 2016; Kierzkowski et al., 2016). Many economists and policy makers

Figure 1: Adjustments of the U.K. economy following the Brexit vote

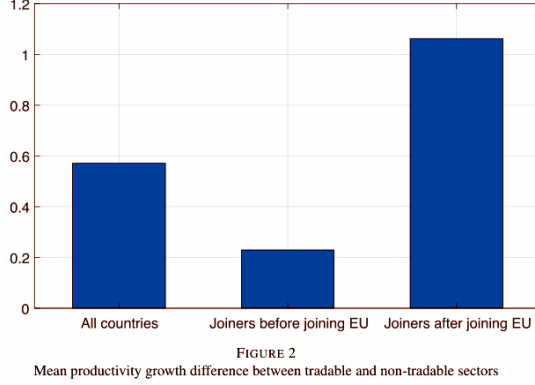
5.2 Figure 2: Mean productivity growth difference between tradable and non-tradable sectors and Table 1: The effects of joining the EU on productivity growth differentials between sectors

Read:

- Figure 2 shows the mean productivity growth difference between tradable and non-tradable sectors.
- The mean difference widens after countries join the EU.
- Table 1 estimates the effect formally using country-sector-quarter regressions.
- The key coefficient is $\text{Tradable} \times \text{Joined EU}$.
- The coefficient is positive and statistically significant across columns, around 0.44–0.48 percentage points per quarter.
- The most demanding specifications include country-by-sector and country-by-time fixed effects.

Interpretation: Figure 2 provides descriptive evidence that tradable productivity growth is more favorable after EU entry. Table 1 shows the result is not driven by simple country averages or global time effects. The coefficient means that after EU entry, tradable-sector productivity growth rises relative to non-tradable-sector productivity growth. In annual terms the effect is large, above 1.6 percentage points. This is not a direct Brexit estimate, but it provides the empirical basis for treating a reduction in EU integration as bad news for tradable productivity growth.

Conclusion: The EU-entry evidence supplies the sign and economic plausibility of the model shock: trade integration is linked to tradable productivity growth.



(a) Figure 2: Mean productivity growth difference between tradable and non-tradable sectors

LHS variable	Growth in real output per hour ($\Delta z_{c,t,t}$)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Tradable $\times$ joined EU	0.443*** [0.117]	0.442*** [0.117]	0.444*** [0.117]	0.441*** [0.118]	0.484*** [0.134]	0.438*** [0.117]	0.482*** [0.134]
Tradable	0.184*** [0.034]	0.184*** [0.034]		0.185*** [0.034]			
Joined EU	0.396** [0.168]	0.137 [0.118]	0.395** [0.168]	0.453** [0.194]	0.139 [0.126]		
Observations	25,006	25,006	25,006	25,006	25,006	25,006	25,006
Country FE	–	✓	–	–	–	–	–
Sector FE	–	–	✓	–	–	✓	–
Time FE	–	–	–	✓	✓	–	–
Country $\times$ sector FE	–	–	–	–	✓	–	✓
Country $\times$ time FE	–	–	–	–	–	✓	✓
R-squared	0.002	0.010	0.005	0.044	0.060	0.238	0.242

Notes: Estimation results of equation (1). The different columns correspond to different fixed effect (FE) specifications. Standard errors are shown in square brackets and are clustered at the country level. *** $p < 0.01$, ** $p < 0.05$.

(b) Table 1: The effects of joining the EU on productivity growth differentials between sectors

5.3 Table 2: Industries shares in the non-tradable and tradable sector (%)

Read:

- Table 2 reports a 2018 snapshot of nominal output shares by broad industry group.
- Services dominate both sectors: 83.26% of non-tradable output and 73.50% of tradable output.
- Manufacturing is much more important in tradables: 21.56% of tradable output versus 1.31% of non-tradable output.
- Construction is assigned to non-tradables and accounts for 11.34% of the non-tradable sector.
- Agriculture, mining, and water-related industries are small shares.

Interpretation: The table prevents a common misunderstanding: tradables are not just manufacturing exports. In the U.K., many tradable activities are services, especially financial and insurance-related services. The model's tradable sector is therefore a broad sector of internationally contestable output, not a narrow goods sector.

Conclusion: The tradable/non-tradable split is economically meaningful for the U.K. because it captures services trade as well as goods trade.

TABLE 2
Industries shares in the non-tradable and tradable sector (%)

	Non-tradable	Tradable
Agriculture	0.05	1.41
Mining and quarrying	0.00	2.60
Manufacturing	1.31	21.56
Electricity, gas, steam air conditioning	2.49	0.00
Water supply, sewage, waste mgmt	1.53	0.93
Construction	11.34	0.00
Services	83.26	73.50

Notes: Nominal output shares of each SIC industry broken down by the classification into tradable and non-tradable sectors. This is shown as a snapshot for the year 2018. The supply-and-use tables are used to calculate the tradability index; nominal GVA data at factor prices are taken from low level aggregates published by the ONS.

Table 2: Industries shares in the non-tradable and tradable sector (%)

5.4 Table 3: Observables used for model estimation and Table 4: Calibrated parameter values

Read:

- Table 3 lists nine observables and their transformations.
- Consumption/GDP and investment/GDP are nominal ratios from 1987:Q3 to 2016:Q2.
- Total hours are scaled by population and demeaned.
- Sectoral labor productivity growth begins in 1994:Q1.
- Relative price data enter at quarterly and annual frequency because different data series have different availability.
- Two real U.S. interest rate measures proxy for the foreign real rate.
- Table 4 reports calibrated parameters: sectoral labor disutility targets, depreciation, consumption/output, trend productivity growth, discounting, and the debt-elastic external premium.

Interpretation: Table 3 maps model objects to data. It explains what the Kalman filter sees during estimation. Table 4 pins down long-run targets that the estimated model must respect. Together, the tables show that the model is not calibrated only to the post-Brexit episode. It is disciplined by a broad pre-referendum macro data set.

Conclusion: The paper’s quantitative exercise rests on a detailed pre-Brexit measurement system, not on fitting the model directly to the post-vote outcomes.

TABLE 3
Observables used for model estimation

Time series	Transformation	Time period
Consumption/GDP	Nominal ratio	1987:Q3–2016:Q2
Investment/GDP	Nominal ratio	1987:Q3–2016:Q2
Total hours (scaled by population)	Dev. from mean	1987:Q3–2016:Q2
Labour productivity in T	% QoQ	1994:Q1–2016:Q2
Labour productivity in N	% QoQ	1994:Q1–2016:Q2
Relative price of non-tradable goods	% QoQ	1997:Q1–2016:Q2
Relative price of non-tradable goods	% Annual	1990–2016
Real U.S. interest rate (based on SPF infl. exp.)	Percent	1987:Q3–2016:Q2
Real U.S. interest rate (based on 5-year inf. MA)	Percent	1987:Q3–2016:Q2

Notes: List of time series used to estimate the model. We use the Kalman filter to address the fact that observables become available at different starting dates. The estimation period ends in 2016:Q2, the quarter of the referendum.

(a) Table 3: Observables used for model estimation

	Description	Source	Period	Value or target
θ_T	Disutility of labour (T)	ONS/own calc.	1994–2016	$n_T/n = 0.45$
θ_N	Disutility of labour (N)	ONS/own calc.	1994–2016	$n_N/n = 0.55$
δ_M	Depreciation in M	ONS	1987–2016	$\frac{\delta}{\gamma} = 0.192$
$\frac{c}{y}$	Consumption/GDP	own calculations	1987–2016	0.644
$\tilde{g}_T$	Trend growth rate of productivity (T)	ONS/own calc.	1994–2016	1.6% ann.
$\tilde{g}_N$	Trend growth rate of productivity (N)	ONS/own calc.	1994–2016	1.2% ann.
β	Discount factor			$r^* = 0.01$
ψ	Debt-elasticity of premium			5×10^{-6}

(b) Table 4: Calibrated parameter values

5.5 Table 5: Priors and posteriors for estimated parameters

Read:

- The posterior mean of the intertemporal elasticity parameter γ remains near 2.00.
- The posterior ratio of capital shares α_T/α_N is about 1.3.
- Labor supply elasticities are estimated at 2.55 for tradables and 2.65 for non-tradables.
- The tradable share ζ rises from prior mean 0.59 to posterior mean 0.62.
- Investment adjustment costs are high: 14.13 in tradables and 12.28 in non-tradables.
- The elasticity-of-substitution parameter σ has posterior mean 0.68, implying gross complementarity between tradables and non-tradables.
- Stationary TFP level shocks are highly persistent, with persistence near 0.90.
- Productivity-growth shock persistence is lower, near 0.23 for tradables and 0.37 for non-tradables.
- Measurement errors are small but statistically relevant for productivity, relative prices, and interest rates.

Interpretation: The posterior estimates support the paper’s mechanism. Complementarity between tradables and non-tradables means that lower expected future productivity reduces demand broadly, not just in the affected sector. High adjustment costs slow capital reallocation and help create persistent dynamics. Persistent TFP level shocks and lower-persistence growth shocks separate cyclical movements from trend-growth news. The estimated tradable share matters because tradable productivity changes have large aggregate effects.

Conclusion: Table 5 gives the model its quantitative behavior: complementarity, slow adjustment, and persistent shocks make the Brexit news shock generate rich sectoral dynamics.

TABLE 5
Priors and posteriors for estimated parameters

Description		Prior			Posterior	
		Prior Distr.	Mean	Std	Mean	90% HPDI
Structural parameters						
γ	Inter-temporal elast. of subs.	Gamma	2	0.1	2.00	1.83 2.16
α_T/α_N	Ratio of capital shares (T over N)	Gaussian	1.3	0.05	1.3	1.22 1.38
ω_T	Elast. of labour supply (T)	Gamma	2	0.2	2.55	2.24 2.86
ω_N	Elast. of labour supply (N)	Gamma	2	0.2	2.65	2.35 2.95
ζ	Tradable share	Beta	0.59	0.01	0.62	0.61 0.63
ϕ_T	Capital adjustment cost in T	Gamma	5	4	14.13	8.02 20.12
ϕ_N	Capital adjustment cost in N	Gamma	5	4	12.28	6.13 18.39
σ	Elast. of subs. between T and N	Gamma	1	0.2	0.68	0.44 0.90
Shocks						
ς_N^g	St.dev. of prod. growth shock in N	Inv. Gamma	0.1	2	0.01	0.013 0.014
ς_T^g	St.dev. of prod. growth shock in T	Inv. Gamma	0.1	2	0.012	0.011 0.014
ς_s	St.dev. of expenditure shock	Inv. Gamma	0.1	2	0.02	0.017 0.022
ς_μ	St.dev. of foreign interest rate shock	Inv. Gamma	0.1	2	0.008	0.007 0.009
ς_ν	St.dev. of preference shock	Inv. Gamma	0.1	2	0.027	0.022 0.033
ς_T^d	St.dev. of TFP level shock in T	Inv. Gamma	0.1	2	0.012	0.011 0.014
ς_N^d	St.dev. of TFP level shock in N	Inv. Gamma	0.1	2	0.012	0.011 0.014
ς_ϵ	St.dev. of labour supply shock	Inv. Gamma	0.1	2	0.017	0.015 0.019
ϱ_N^g	Persistence of prod. growth shock in N	Beta	0.5	0.15	0.37	0.18 0.56
ϱ_T^g	Persistence of prod. growth shock in T	Beta	0.5	0.15	0.23	0.12 0.35
ϱ_s	Persistence of expenditure shock	Beta	0.5	0.15	0.58	0.35 0.82
ϱ_μ	Persistence of foreign interest rate shock	Beta	0.5	0.15	0.56	0.46 0.65
ϱ_ν	Persistence of preference shock	Beta	0.5	0.15	0.92	0.88 0.98
ϱ_N^d	Persistence of TFP level shock in N	Beta	0.5	0.15	0.91	0.82 0.98
ϱ_T^d	Persistence of TFP level shock in T	Beta	0.5	0.15	0.90	0.85 0.97
ϱ_ϵ	Persistence of labour supply shock	Beta	0.5	0.15	0.95	0.91 0.99
Measurement errors						
ι_P	Relative price (annual)	Inv. Gamma	0.01	0.0001	0.01	0.01 0.01
ι_N	Labour productivity in N	Inv. Gamma	0.01	0.005	0.005	0.004 0.007
ι_T	Labour productivity in T	Inv. Gamma	0.01	0.005	0.006	0.005 0.008
ι_{R_1}	Real foreign interest rate (1)	Inv. Gamma	0.01	0.01	0.002	0.002 0.002
ι_{R_2}	Real foreign interest rate (2)	Inv. Gamma	0.01	0.01	0.002	0.002 0.002

Table 5: Priors and posteriors for estimated parameters

5.6 Figure 3: Main model responses in benchmark Brexit scenario and Figure 4: Additional model responses in benchmark Brexit scenario

Read:

- Figure 3 Panel A shows the anticipated path of lower future tradable productivity growth.
- Figure 3 Panel B shows GDP falling gradually relative to the no-Brexit path.
- Figure 3 Panel C shows tradable output temporarily rising during the anticipation phase.
- Figure 3 Panel D shows non-tradable output falling.
- Figure 3 Panel E shows the relative price of non-tradables falling immediately.
- Figure 3 Panel F shows the domestic interest-rate spread falling.
- Figure 3 Panels G and H show investment falling and aggregate hours remaining comparatively stable early on.
- Figure 4 adds the trade balance, consumption, sectoral consumption, sectoral hours, and sectoral wages.
- Figure 4 shows that both consumption components fall, tradable hours rise temporarily, non-tradable hours fall, and wages decline in both sectors over time.

Interpretation: Figures 3 and 4 are the structural heart of the paper. The model produces a sequence of events: news arrives, relative prices adjust immediately, current tradable production enters a favorable window, non-tradables contract, and aggregate demand weakens because permanent income is lower. Once the future productivity decline arrives, the short-run tradable advantage disappears and the long-run cost

dominates. The figures also show why the mechanism is not just a substitution effect. A permanent-income effect lowers consumption and investment across sectors.

Conclusion: The model converts a future productivity-growth loss into immediate macroeconomic adjustment through relative prices, expectations, and intertemporal choices.

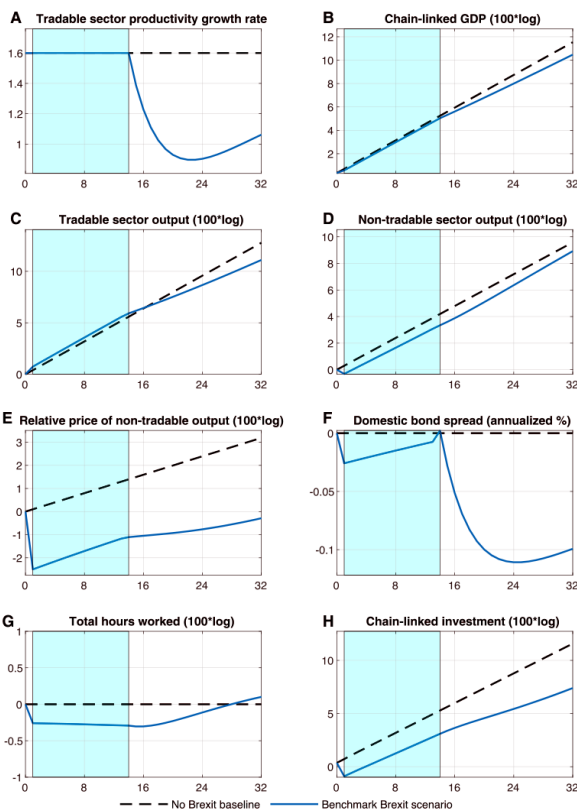


FIGURE 3
Main model responses in benchmark Brexit scenario

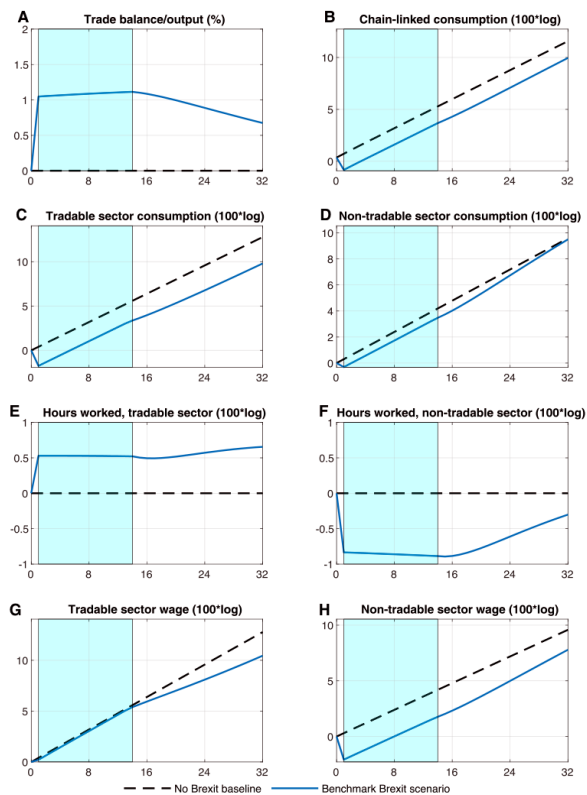


FIGURE 4
Additional model responses in benchmark Brexit scenario

(a) Figure 3: Main model responses in benchmark Brexit scenario

Panel A of Figure 3 shows the trajectory of tradable sector productivity growth that we fee
(b) Figure 4: Additional model responses in benchmark Brexit scenario

5.7 Table 6: Comparison of Brexit simulation with U.K. data

Read:

- Table 6 compares simulation and data at 6-quarter and 12-quarter horizons.
- Tradable output rises in both simulation and data: at 12 quarters, the simulation explains about 70% of the data movement.
- Non-tradable output falls in both simulation and data: at 12 quarters, the simulation explains about 30% of the data movement.
- Tradable hours rise in both simulation and data, but the observed 12-quarter increase is much larger than in the model.
- Non-tradable hours fall in both simulation and data; the model somewhat overpredicts the decline.
- Tradable consumption at 12 quarters is matched closely: both simulation and data are around -2.2.

- Non-tradable consumption falls in both simulation and data; the model explains roughly 60% of the 12-quarter decline.
- Total consumption is captured well: the simulation explains around 80% of the 12-quarter data movement.
- Investment falls in both simulation and data, but the observed decline is larger and more delayed.
- The trade-balance comparison is weaker: the model predicts an improvement, while the 12-quarter data response is near zero or slightly negative.
- Average real wages fall in both simulation and data, though the data decline is roughly twice as large.

Interpretation: Table 6 is the paper’s main model-validation table. It shows that the model matches signs for most sectoral and aggregate variables. The strongest successes are sectoral output directions, consumption, and wages. The largest shortcomings are investment timing and trade-balance magnitude. This pattern is informative: the productivity-news mechanism explains first-moment reallocation and permanent-income responses, while uncertainty and inventory/trade frictions likely explain what the model misses.

Conclusion: The benchmark news shock explains a meaningful share of post-Brexit adjustment, but not the full event. The table identifies where additional mechanisms are needed.

TABLE 6
Comparison of Brexit simulation with U.K. data

	6 quarter horizon			12 quarter horizon		
	Simulation	Data	Ratio	Simulation	Data	Ratio
Sectoral variables						
Tradable output	0.4	1.0	0.4	0.4	0.5	0.7
Non-tradable output	−0.7	−1.3	0.6	−0.8	−2.3	0.3
Tradable sector hours	0.5	1.1	0.5	0.5	5.4	0.1
Non-tradable sector hours	−0.9	−0.4	2.0	−0.9	−0.6	1.6
Tradable consumption	−2.2	−0.5	4.1	−2.2	−2.2	1.0
Non-tradable consumption	−0.6	−1.8	0.4	−0.7	−1.1	0.6
Aggregate variables						
GDP	−0.2	−0.1	1.9	−0.2	−0.7	0.3
Total consumption	−1.4	−1.2	1.2	−1.4	−1.7	0.8
Total investment	−1.7	−0.5	3.1	−1.9	−4.2	0.5
Tradable net export ratio	1.1	0.5	2.2	1.1	−0.1	−18.1
Average real wage	−1.2	−2.5	0.5	−1.2	−2.8	0.4

Table 6: Comparison of Brexit simulation with U.K. data

5.8 Table 7: Estimates of long-run effects of Brexit on U.K. GDP

Read:

- Table 7 compares the paper’s long-run GDP loss with estimates from the Brexit literature.
- Ebell and Warren estimate 2.7–3.7%.
- IMF estimates 5.2–7.8%.
- Kierzenkowski et al. estimate 2.7–7.5%.
- The U.K. Government estimates 6.3–10.7%.
- The paper’s simulation implies a 3.6% long-run GDP reduction.

Interpretation: The paper’s estimate is near the lower end of the literature because the model isolates one mechanism: lower future tradable productivity growth. Other analyses often incorporate broader policy changes, trade costs, migration, regulatory divergence, or uncertainty. The table therefore helps calibrate the scope of the paper’s claim.

Conclusion: The model’s 3.6% long-run GDP loss is economically meaningful but conservative relative to many broader Brexit estimates.

TABLE 7
Estimates of long-run effects of Brexit on U.K. GDP

Study	Estimated reduction in GDP (%)
Ebell and Warren (2016)	2.7–3.7
IMF (2018)	5.2–7.8
Kierzenkowski <i>et al.</i> (2016)	2.7–7.5
UK Government (2018)	6.3–10.7
Our simulation	3.6

Table 7: Estimates of long-run effects of Brexit on U.K. GDP

6 Interpretive synthesis and limitations

- **Why tradables expand after bad news.** The productivity decline is expected in the future. Current tradable productivity has not yet fallen, while relative prices adjust immediately, creating a temporary advantage for tradable producers.
- **Why non-tradables contract.** The relative price of non-tradables falls and lower expected lifetime income reduces domestic demand. Non-tradables cannot redirect output to export markets as easily.
- **Why GDP still slows.** Tradables are only part of the economy, and households and firms cut spending when expected future income declines.
- **What the model misses.** The investment shortfall and trade-balance dynamics likely require additional mechanisms, especially uncertainty, inventories, gross trade flows, imported-input dynamics, and monetary-financial channels.
- **Alternative shocks.** Level shocks and non-tradable shocks fail on important margins, especially interest-rate persistence and long-run GDP implications.

Interpretation: The mechanism is strong but not exhaustive. It is best read as a disciplined account of the productivity-news channel, not a complete model of every Brexit adjustment margin.

7 Short summary

- The paper studies U.K. macroeconomic adjustment after the 2016 Brexit referendum.
- The key facts are tradable-sector resilience, non-tradable stagnation, falling relative prices of non-tradables, and slower aggregate demand.
- The authors construct new U.K. quarterly sectoral data using 2-digit SIC industry information.
- Cross-country evidence shows that EU entry is associated with faster tradable productivity growth relative to non-tradables.
- The model is a two-sector real small open economy with sectoral productivity trends, CES consumption and investment, sector-specific labor, external borrowing, and capital adjustment costs.
- The Brexit vote is modeled as news about a future decline in tradable productivity growth.
- The shock is not estimated from post-Brexit data; parameters are estimated on pre-referendum data.
- The simulation matches the sign of most post-referendum variables and explains a meaningful share of the magnitudes.
- The model implies a long-run GDP loss of about 3.6%.

- Remaining discrepancies point to uncertainty, investment option value, gross-trade dynamics, inventories, and financial channels.

8 Conclusion

8.1 Core conclusion

The Brexit vote can be understood as news about lower future tradable productivity growth. This explains why tradable producers did not collapse immediately and why non-tradables absorbed much of the short-run adjustment.

8.2 Economic intuition

Expectations move before realized fundamentals. Once agents expect lower future tradable productivity, the relative price of non-tradables falls, shifting activity toward tradables in the short run. At the same time, expected lifetime income falls, so consumption and investment decline.

8.3 Final takeaway

The paper does not claim that productivity news is the only Brexit mechanism. It shows that this one mechanism explains a surprisingly rich set of facts: falling non-tradable relative prices, resilient tradables, weak non-tradables, lower spending, and a lower long-run GDP path. The remaining gaps point to uncertainty, gross trade, inventories, and financial channels.